

Heart Disease Prediction System

Developer Notes

Overview

Built as an end-to-end data science project to demonstrate the complete machine learning lifecycle, including data preprocessing, visualization, model training, deployment, and version control.

Developer Notes

- Focused on building a clean ETL pipeline, an interactive Power BI dashboard, and a deployable Streamlit application.
- Random Forest was selected as the final model after comparing multiple algorithms based on overall performance.
- The Random Forest model (200 decision trees) produced prediction scores that typically ranged from 76% to 86% on this dataset. To make the output easier for non-technical users to interpret, the application rescales the displayed confidence score to a 0–100% range for visualization only. This does not change the model's underlying prediction or its performance.
- Future improvements include using a larger dataset, hyperparameter tuning, cross-validation, model calibration, explainable AI (SHAP), and database integration with user authentication.